A Comparative Study of Three Sampling Methods in Investigating the Average Amount of Money Spent by University Students on Online Shopping in One Year

1. Statement of objectives

The primary aim of this study was to compare the accuracy and factual conformity of three distinct sampling methods (random, stratified, and systematic) in estimating and analyzing the per capita consumption of undergraduate students at Jinan University over one year. To achieve this objective, questionnaires and sampling statistics were utilized to estimate and analyze the per capita consumption. Through an online questionnaire platform, undergraduate students were invited to report their expenditures on various goods based on their actual circumstances. The questionnaire incorporated demographic information for subsequent sampling.

Given the complexity of the statistical work, we do not conduct a full analysis of the 20,000 questionnaires but use a sampling approach for estimation. In the data processing and analysis stage, the valid questionnaires received were categorized according to grade and gender, and three methods were used to select 500 samples from which to analyse to derive the amount spent by undergraduate students at Jinan University in one year under different sampling methods. By comparing and analysing the results of the three methods, we assessed the accuracy and factual conformity of the various sampling methods in estimating and analyzing the amount spent by undergraduate students.

Through this study, we expect to provide researchers and policymakers with references on sampling method selection and data analysis to facilitate more accurate and reliable data estimation and analysis.

2. Target population

The target population for this survey is the 20,000 undergraduate students at Jinan University. (This is the design for an ideal survey situation)

3. The frame

The sampling for this survey was based on the sum of the dollar amounts of various types of consumer goods, and the sampling frame for undergraduate students at Jinan University corresponded to a dataset of total online shopping spending over one year. The survey was conducted through an online questionnaire platform that provided links to the target population, which included undergraduate students at Jinan University. The questionnaire collected data on the amount spent on different types of consumer goods and also included personal information such as gender and year. The sampling unit for this survey was the sum of the consumption amounts of various goods, representing the total expenditure of undergraduate students at Jinan University in a year.

To process and analyse the data, valid questionnaires were categorised by grade and gender. Three sampling methods, namely random sampling, stratified sampling and systematic sampling, were used to draw 500 samples from the population dataset for analysis respectively.

4. Sample method

Simple random sampling: a random sample drawn from the total population, where each sample has an equal chance of being drawn and the samples are independent of each other. In this study, the simple random sampling method was first chosen and 500 sample data were drawn from 20,000 data. In the sampling process, each data has an equal probability of p_i being selected and the samples were independent of each other. Here, $p_i = \frac{1}{N}$, where p_i denotes the probability of each data being selected and N denotes the population size. The analysis of the sample data yields a sample mean of $\bar{y}$. We can use the sample mean to estimate the mean of population $\hat{\bar{Y}}$. This gives us an estimate of the per ca pita consumption of 20,000 people. Finally, we analyse and evaluate the estimation results. Specific steps:

a. A random generator in R was used to generate 500 random integers between 1 and 20,000 to be used as the numbers of the sampled units, and then the corresponding numbered units were selected from the population to form the sample. Find the sample mean $\bar{y}$ and obtain an estimate of the population

mean$\widehat{\bar{Y}}$. $\widehat{\bar{Y}} = \bar{y}$.

b. The estimated value error is:$\left|\widehat{\bar{Y}} - \bar{y}\right|$.

c. Confidence interval calculation: $\left[\bar{y} \pm Z_{\frac{\alpha}{2}} \times \frac{s}{\sqrt{n}}\right]$, where $s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2}$, we set the next calculation to be calculated at a 95% confidence level, therefore, $Z_{\frac{\alpha}{2}}$ =1.96.

Stratified sampling is a probability sampling method in which the total population is divided into a number of non-overlapping sub-totals (called strata) and then a simple random sample is drawn independently from each sub-total.Specific steps:

a. In this survey, we can perform stratified sampling based on gender and grade. First, we need to calculate the sampling proportion for each level. Here we use equal proportional stratified sampling, i.e. each level is sampled in the same proportion as its share of the total. It is known that there are two genders (male and female) and four grades (1, 2, 3 and 4), for a total of eight levels. Next, we will calculate the sample size for each level and draw the appropriate number of samples from each level.

b. According to the definition of stratified sampling, each level is sampled in the same proportion as its share of the population, so that the sample means at each level can be weighted equally to estimate the population mean. Specifically, let the i the sample size of the stratum be ni and the sample mean be $\bar{y}_i$ andi The proportion of level 1 in the population isp_i , then the estimate of the population mean is $\hat{\bar{y}} = \sum_{i=1}^{8} p_i \bar{y}_i$, where $p_i = \frac{N_i}{N}$, where N is 20, 000 and N_i is 2,500 and p_i is $\frac{1}{8}$.

c. The estimated value error is: $\left|\widehat{\bar{Y}} - \bar{y}\right|$.

d. Since we already know the proportion of each level in the population, confidence intervals can be calculated based on a weighted average approach. Specifically, the confidence intervals for the estimates are

$\left[\bar{y} \pm Z_{\frac{\alpha}{2}} \times \frac{\sqrt{\sum_{i=1}^{8} \frac{N_i}{N} s_i^2}}{\sqrt{n}}\right]$, wheres_i^2 is the variance at level i.$Z_{\frac{\alpha}{2}}$ = 1.96.

Systematic sampling is a probability sampling method which draws samples from the total population at fixed intervals. Specific steps:

a. In this survey, the population size is 20,000 (i.e. 20,000 questionnaires) and the sample size is 500. Calculate the sampling interval (k): the sampling interval is the number of elements you skip each time you take a sample from the population. The formula for calculating this is $k = \frac{N}{n}$. In this example, the sampling interval is $k = 20{,}000 / 500 = 40$. Choose a random starting point (s): choose a random integer between 1 and 40 as the starting point. Samples are drawn at the sampling interval: from the starting point (s), a sample is drawn every 40 elements until 500 samples are drawn.

As systematic sampling is based on the same formula as simple random sampling, the process of step b, c and d will not be repeated.

5. Method of measurement and implementation

A questionnaire was used as the primary method of data collection for this study. Rigorous measures were taken at the questionnaire design stage to minimise errors, including careful content design, selection of appropriate question formats and inclusion of an optimal number of questions to address issues such as unclear presentation and respondent misunderstanding. Mandatory questions and verification of identifying information were also included to ensure the accuracy of responses.

This study aims to compare the accuracy of the three sampling methods by comparing the estimates of the three sampling methods with the mean of a population data set consisting of 20,000 data points.

Throughout the study, careful attention will be paid to the randomness and representativeness of the sample selection to ensure the credibility of the results. Statistical analysis techniques, such as error calculations and confidence interval estimation, will be used to assess the validity of the sampling methods and to determine the margin of error.

6. The pretest

Ten undergraduate questionnaires from Jinan University were randomly selected

as pretest data to assess the feasibility and completion of this survey questionnaire and to adjust and optimise the questions. The results of the pretest showed that the questionnaire design of this survey was accurate and scientific, logical and easy to understand.

7. Data simulation

As we were unable to obtain the real consumption data of the undergraduate population at Jinan University, the population data for this survey were randomly generated using R language to simulate the data. In generating the data, we set the ratio of male to female to 1:1, and the ratio of freshman:sophomore:junior: senior to 1:1:1:1 according to the basic situation of Jinan University. At the same time, we randomly selected four options for each person's specific consumption amount and summed these amounts to arrive at the total consumption amount for each person. (The code is attached)

8. Calculation and analysis

Population: The per ca pita consumption of the population $\bar{Y} = \sum_{i=1}^{N} y_i =$ 2475.33, standard deviation of the population $\sigma = \sqrt{\frac{\sum_{i=1}^{N} (y_i - \bar{Y})^2}{N}}$ =638.09

Random sampling: The per ca pita consumption of the sample based on a random sample $\bar{y}_{SRS}$ is 2455.80 and the standard deviation of the sample σ_{SRS} is 621.71. Based on the formula, the per ca pita consumption of the population is estimated to be $\hat{\bar{Y}}_{SRS} = \bar{y}_{SRS}$ = 2455.80. The error is $\left|\bar{Y} - \hat{\bar{Y}}_{SRS}\right|$ =19.93, with a confidence interval $\left[\bar{y} \pm Z_{\frac{\alpha}{2}} \times \frac{s}{\sqrt{n}}\right]$ is $[2400.70, 2510.90]$.

Stratified sampling: The per ca pita consumption of the sample based on the stratified sample $\bar{y}_{Stratified}$ is 2493.00 and the standard deviation of the sample $\sigma_{Stratified}$ is 630.91. Given that the stratification is based on gender and grade level, the sampling proportion is uniform across each stratum. Consequently, the stratified sample mean can be used to accurately represent the population mean, and according to the formula, the per ca pita consumption of the population is estimated to be $\hat{\bar{Y}}_{Stratified} = \bar{y}_{Stratified}$ = 2493.00. The error is $\left|\bar{Y} - \hat{\bar{Y}}_{Stratified}\right|$ = 17.27, and the weighted confidence interval is calculated $\left[\bar{y} \pm Z_{\frac{\alpha}{2}} \times \frac{\sqrt{\sum_{i=1}^{8} \frac{N_i}{N} s_i^2}}{\sqrt{n}}\right] = [2484.95, 2501.05]$.

Systematic sampling: The per capita consumption of the sample based on a random sample $\bar{y}_{Systematic}$ is 2512.80 and the standard deviation of the sample $\sigma_{Systematic}$ is

633.81. Based on the formula, the per capita consumption of the population is estimated to be $\widehat{\bar{Y}}_{Systematic} = \bar{y}_{Systematic}$ = 2512.80. The error is $\left|\bar{Y} - \widehat{\bar{Y}}_{SRS}\right|$ = 37.07, with a confidence interval $\left[\bar{y} \pm Z_{\frac{\alpha}{2}} \times \frac{s}{\sqrt{n}}\right]$ is [2504.66,2520.94].

	Population mean estimate	Error	Confidence interval
Simple Random Sampling	2455.80	19.93	[2447.89,2463.71]
Stratified Sampling	2493.00	17.27	[2484.95,2501.05]
Systematic sampling	2512.80	37.07	[2504.66,2520.94]

Table 1

Through analysis and comparison of Table 1, it can be observed that stratified sampling exhibits the smallest absolute error, while systematic sampling exhibits the largest. Consequently, the results indicate that stratified sampling is the most effective method for this survey.

9. Overall summary and conclusion

A comprehensive analysis revealed that the stratified systematic sampling method yielded the most accurate sample results.

In practice, it would have been more reasonable to distribute the personal information survey link to 20,000 Jinan University undergraduates and collect personal information such as gender and year. Alternatively, personal information could be collected through the school's roster, followed by sampling and administering the questionnaire to 500 people in the sample. These methods would be more suitable if the population size was larger.

It is important to note that this survey is only intended to compare the effects of different sampling methods. In actual surveys, the required sample size should be estimated and the appropriate sampling method selected based on the required level of accuracy. Furthermore, the performance of the three sampling methods is based solely on the simulated questionnaire and data set, and may not fully reflect their effectiveness in practical applications. Therefore, the selection of the appropriate sampling method should take into account the actual situation and needs of the study.

Appendix

Jinan University Annual Shopping Festival Undergraduate Online Consumption Amount Survey

Dear Jinan University undergraduates, hello! In order to better understand the consumption behavior and the consumption amount of Jinan University undergraduates , we have developed the following questionnaire. Please answer the questions honestly and carefully based on your actual situation. Thank you for your cooperation!

I. Basic Information

1. Gender:

( ) Male ( ) Female

2. Grade:

( ) Freshman ( ) Sophomore ( ) Junior ( ) Senior

II. Online Consumption Situation

3. During the year-long online shopping period, your consumption amount for clothing products was:

( ) 0 yuan ( ) 100 yuan ( ) 300 yuan ( ) 500 yuan

4. During the year-long online shopping period, your consumption amount for shoes and hats products was:

( ) 0 yuan ( ) 100 yuan ( ) 300 yuan ( ) 500 yuan

5. During the year-long online shopping period, your consumption amount for electronic products was:

( ) 0 yuan ( ) 100 yuan ( ) 300 yuan ( ) 500 yuan

6. During the year-long online shopping period, your consumption amount for mobile phones and accessories was:

( ) 0 yuan ( ) 100 yuan ( ) 300 yuan ( ) 500 yuan

7. During the year-long online shopping period, your consumption amount for cosmetics and skincare products was:

( ) 0 yuan ( ) 100 yuan ( ) 300 yuan ( ) 500 yuan

8. During the year-long online shopping period, your consumption amount for sports and outdoor products was:

( ) 0 yuan ( ) 100 yuan ( ) 300 yuan ( ) 500 yuan

9. During the year-long online shopping period, your consumption amount for books was:

( ) 0 yuan

( ) 100 yuan

( ) 300 yuan

( ) 500 yuan

10. During the year-long online shopping period, your consumption amount for stationery products was:

( ) 0 yuan ( ) 100 yuan ( ) 300 yuan ( ) 500 yuan

11. During the year-long online shopping period, your consumption amount for daily necessities was:

( ) 0 yuan ( ) 100 yuan ( ) 300 yuan ( ) 500 yuan

12. During the year-long online shopping period, your consumption amount for food and beverage products was:

( ) 0 yuan ( ) 100 yuan ( ) 300 yuan ( ) 500 yuan

13. During the year-long online shopping period, your consumption amount for home textiles and home products was:

( ) 0 yuan ( ) 100 yuan ( ) 300 yuan ( ) 500 yuan

Thank you for your patience in completing this survey! With your support, our research will be more valuable.